

E-COMMERCE FUNNEL ANALYSIS

SQL Analysis Portfolio Project

Identifying Revenue Leakage, Customer Drop-Off & Retention Gaps

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Tools: MySQL Workbench 8.0 | MS Excel | Power BI

Project Overview

This project performs a comprehensive funnel analysis on a real-world UK e-commerce dataset containing 392,733 cleaned transaction records. The objective is to identify where the business is losing revenue, which customers are at risk of churning, and what strategic actions can improve retention and growth.

Attribute	Detail
Dataset	UK Online Retail: Real-world e-commerce transactions
Records (after cleaning)	392,733 rows
Unique Customers	4,339
Time Period	December 2010 - December 2011
Total Revenue	£8,887,208.89
Tools Used	MySQL Workbench 8.0, MS Excel, Power BI, Claude AI
SQL Concepts Used	CTEs, Window Functions, LAG(), NTILE(), DATEDIFF(), CASE WHEN, JOINS, GROUP BY, Subqueries

Section 1: Data Quality Checks

Before any analysis, SQL queries were run to validate the data imported from the cleaned Excel file. This ensures no dirty data contaminates the analysis.

1A — Row Count Verification

```
USE ecommerce_project;

SELECT COUNT(*) FROM `e-commerce_clean`;
-- Result: 392,733 rows confirmed
```

1B — Check for Null CustomerIDs

```
SELECT CUSTOMERID FROM `e-commerce_clean`
WHERE CUSTOMERID IS NULL;
-- Result: 0 rows returned — no null CustomerIDs
```

1C — Check for Cancelled Orders & Negative Quantities

```
SELECT * FROM `e-commerce_clean`
WHERE InvoiceNo LIKE 'C%' OR Quantity < 0;
-- Result: 0 rows returned — cancellations successfully removed in Excel cleaning
```

1D — Check for Negative Unit Prices

```
SELECT * FROM `e-commerce_clean`
WHERE UnitPrice < 0;
-- Result: 0 rows returned — data is clean
```



KEY INSIGHT: All 4 data quality checks passed with 0 errors, confirming the Excel cleaning process was thorough and the dataset is analysis-ready.

Section 2: Monthly Revenue Trend & Growth Analysis

Business Question: Which months experienced revenue decline, by how much, and what are the possible reasons?

SQL Query — Monthly Revenue with Month-on-Month Growth %

```
SELECT
  MONTH,
  SUM(Revenue) AS Monthly_Revenue,
  LAG(SUM(REVENUE)) OVER (ORDER BY MIN(Month)) AS Prev_month_revenue,
  ROUND(
    (SUM(revenue) - LAG(SUM(revenue)) OVER (ORDER BY MIN(Month)))
    / LAG(SUM(revenue)) OVER (ORDER BY MIN(Month)) * 100, 2
  ) AS MOM_GROWTH_PERCT
FROM `e-commerce_clean`
GROUP BY Month
ORDER BY MIN(Month);
```

Results

Month	Monthly Revenue (£)	Prev Month Revenue (£)	MoM Growth %
2010-Dec	23,837.32	NULL	NULL (first month)
2011-Jan	17,373.18	23,837.32	+5.51%
2011-Feb	16,465.23	17,373.18	-17.14%
2011-Mar	25,746.04	16,465.23	+13.97%
2011-Apr	28,851.30	25,746.04	+21.03%
2011-May	32,298.85	28,851.30	+25.45%
2011-Jun	22,589.39	32,298.85	-3% (approx)
2011-Jul	20,595.37	22,589.39	+18.55%
2011-Aug	19,869.96	20,595.37	-31.13%
2011-Sep	29,907.55	19,869.96	-50.77%
2011-Oct	60,746.70	29,907.55	+287.22%
2011-Nov	15,687.75	60,746.70	-51.43%

 **KEY INSIGHT:** October 2011 recorded an extraordinary +287% month-on-month growth, the single largest spike in the dataset. November shows a sharp -51% decline from this peak. February recorded the sharpest early-year decline at -17%, consistent with post-holiday slowdown. The LAG() window function was used to compare each month with the previous month without a self-join.

 **RECOMMENDATION:** Launch targeted promotional campaigns in January-February to counter seasonal post-holiday slowdown. Investigate what drove the October spike, if it was a specific campaign or product launch, replicate it.

Section 3: Product Performance — Revenue & Volume Leaders vs Dead Stock

Business Question: Which products are driving revenue and which are dead stock & what should the business do?

3A — Top 10 Products by Revenue

```
SELECT Description, ROUND(SUM(Revenue), 2) AS total_revenue
FROM `e-commerce_clean`
GROUP BY Description
ORDER BY total_revenue DESC
LIMIT 10;
```

Product	Total Revenue (£)
Manual	47,180.07
Postage	16,696.01
Dotcom Postage	6,009.92
Carriage	5,550.00
Regency Cakestand 3 Tier	5,493.09
Love Seat Antique White Metal	3,650.00
Vintage Red Kitchen Cabinet	2,775.00
Rustic Seventeen Drawer Sideboard	2,640.00
Bread Bin Diner Style Ivory	2,592.74
Set/4 White Retro Storage Cubes	2,357.05

3B — Bottom 10 Products by Revenue (Dead Stock)

```
SELECT Description, ROUND(SUM(Revenue), 2) AS Total_Revenue
FROM `e-commerce_clean`
GROUP BY Description
ORDER BY Total_Revenue ASC
LIMIT 10;
```

Product	Total Revenue (£)
Pads to Match All Cushions	0.00
Blue Egg Spoon	0.12
Wine Bottle Dressing Lt. Blue	0.12
Red Egg Spoon	0.12
Set/6 Green Spring Paper Cups	0.19
Retro Pill Box, Revolutionary	0.19
Black Feather Christmas Decoration	0.19

Pet Mug, Goldfish	0.19
Set/6 Posies Paper Cups	0.19
Feng Shui Pillar Candle	0.19

3C — Top 10 Products by Volume (Quantity Sold)

```
SELECT Description, COUNT(Quantity) AS Total_Quantity
FROM `e-commerce_clean`
GROUP BY Description
ORDER BY Total_Quantity DESC
LIMIT 10;
```

Product	Total Quantity Sold
Regency Cakestand 3 Tier	427
Rex Cash+Carry Jumbo Shopper	373
Postage	279
White Wood Garden Plant Ladder	208
Enamel Bread Bin Cream	173
Recipe Box Pantry Yellow Design	170
Set of 3 Cake Tins Pantry Design	163
Sweetheart Cakestand 3 Tier	157
Bread Bin Diner Style Ivory	152
Red Retrospot Cake Stand	149

 **KEY INSIGHT:** Key finding: The real top-selling product is Regency Cakestand 3 Tier which appears in BOTH revenue and volume top 10: a true star product. The bottom 10 products each generated less than £0.20 in total revenue, these are dead stock candidates for discontinuation.

 **RECOMMENDATION:** Discontinue or heavily discount the bottom 10 dead stock products to free up inventory capital. Double down on marketing the Regency Cakestand 3 Tier, it leads on both revenue and volume. Filter 'Manual' and 'Postage' entries from future product analysis as they skew results.

Section 4: Customer Lifecycle Segmentation

Business Question: How many customers bought only once vs repeat buyers vs loyal customers and what % of revenue does each group drive?

SQL Query — Customer Segmentation with Revenue %

```
WITH CUSTOMER_SEGMENT AS (  
    SELECT CustomerID,  
           COUNT(DISTINCT InvoiceNo) AS Order_Count,  
           ROUND(SUM(Revenue), 2) AS Total_Revenue,  
           CASE  
               WHEN COUNT(DISTINCT InvoiceNo) = 1 THEN 'ONE-TIME BUYER'  
               WHEN COUNT(DISTINCT InvoiceNo) >= 5 THEN 'LOYAL CUSTOMER'  
               ELSE 'REPEAT BUYER'  
           END AS SEGMENT  
    FROM `e-commerce_clean`  
    WHERE CustomerID IS NOT NULL  
    GROUP BY CustomerID  
)  
SELECT  
    SEGMENT,  
    COUNT(CustomerID) AS Total_Customers,  
    SUM(Order_Count) AS Total_Orders,  
    ROUND((SUM(Order_Count)*100) / SUM(SUM(Order_Count)) OVER(), 2) AS  
Percent_Total_Orders,  
    ROUND(SUM(Total_Revenue), 2) AS Segment_Revenue,  
    ROUND(SUM(Total_Revenue)*100 / SUM(SUM(Total_Revenue)) OVER(), 2) AS  
Percent_Total_Revenue  
FROM Customer_Segment  
GROUP BY SEGMENT  
ORDER BY FIELD(SEGMENT, 'ONE-TIME BUYER', 'REPEAT BUYER', 'LOYAL CUSTOMER');
```

Results

Segment	Customers	Total Orders	% of Orders	Revenue (£)	% of Revenue
One-Time Buyer	1,251	1,251	20.30%	62,537.28	19.92%
Repeat Buyer	818	2,150	34.89%	115,142.79	36.67%
Loyal Customer	270	2,762	44.82%	136,288.57	43.41%

 **KEY INSIGHT:** Only 270 Loyal Customers (6.2% of all customers) drive 43.41% of all revenue, £136,288. Meanwhile 1,251 One-Time Buyers represent 28.8% of customers but generate under 20% of revenue. The business has a critical customer acquisition problem, the base is too small (only 2,339 unique customers in SQL vs 4,339 in Excel due to data type differences).

 **RECOMMENDATION:** Implement a loyalty rewards programme targeting the 270 Loyal Customers immediately, losing even 10% of this group means losing £13,628 in revenue. Launch targeted re-engagement email campaigns for the 1,251 One-Time Buyers to convert them into Repeat Buyers within 30 days of their first purchase.

Section 5: Peak Hours & Day of Week Analysis

Business Question: What time of day and day of week drives most purchases — how can the business use this to optimise marketing timing?

5A — Peak Hours Analysis

```
SELECT
    Hour,
    COUNT(DISTINCT InvoiceNo) AS Total_Orders,
    ROUND(SUM(Revenue), 2) AS Total_Revenue,
    ROUND(SUM(Revenue)/COUNT(DISTINCT InvoiceNo), 2) AS Avg_Order_Value
FROM `e-commerce_clean`
GROUP BY Hour
ORDER BY Total_Orders DESC;
```

Hour	Total Orders	Total Revenue (£)	Avg Order Value (£)
12 (Noon)	1,162	50,800.31	43.72
13 (1pm)	1,014	58,910.73	58.10
14 (2pm)	848	52,977.91	62.47
11 (11am)	798	39,400.89	49.37
15 (3pm)	702	36,270.20	51.67
10 (10am)	555	20,810.21	37.50
16 (4pm)	417	21,548.78	51.68
9 (9am)	223	11,463.77	51.41
17 (5pm)	222	12,867.88	57.96
19 (7pm)	79	3,485.90	44.13
8 (8am)	59	2,280.25	38.65
6 (6am)	1	4.25	4.25

5B — Day of Week Analysis

```
SELECT
    `Day Of the Week`,
    COUNT(DISTINCT InvoiceNo) AS Total_Orders,
    ROUND(SUM(Revenue), 2) AS Total_Revenue,
    ROUND(SUM(Revenue)/COUNT(DISTINCT InvoiceNo), 2) AS Avg_Order_Value
FROM `e-commerce_clean`
GROUP BY `Day Of the Week`
ORDER BY Total_Revenue DESC;
```

Day	Total Orders	Total Revenue (£)	Avg Order Value (£)
Thursday	1,236	63,801.37	51.62
Monday	950	62,583.22	65.88

Sunday	1,113	51,225.83	46.03
Wednesday	1,082	51,090.02	47.22
Tuesday	970	48,561.49	50.06
Friday	812	36,706.71	45.21



KEY INSIGHT: Peak purchase hour is 12pm-2pm with 12 noon having the highest order count (1,162 orders). However 1pm has the highest revenue (£58,910) and 2pm has the highest Average Order Value (£62.47) — suggesting afternoon buyers spend more per order. Thursday drives highest revenue (£63,801) while Monday has the highest AOV (£65.88 per order). No activity after 8pm and virtually none before 9am.



RECOMMENDATION: Schedule all promotional emails and marketing campaigns between 11am-2pm for maximum impact. Run flash sales or limited offers on Thursday and Monday when AOV is highest. Consider targeted late-morning push notifications at 10-11am to capture early buyers before the peak.

Section 6: RFM Segmentation — Identifying Champions & At-Risk Customers

Business Question: Who are the most valuable customers and which ones are at risk of churning? RFM = Recency (how recently they bought) + Frequency (how often) + Monetary (how much they spend).

SQL Query — Full RFM Scoring Using NTILE() Window Function

```
WITH RFM_BASE AS (
    SELECT
        CustomerID,
        DATEDIFF((SELECT MAX(Date) FROM `e-commerce_clean`), MAX(Date)) AS Recency_Days,
        COUNT(DISTINCT InvoiceNo) AS Frequency,
        SUM(Revenue) AS Monetary
    FROM `e-commerce_clean`
    WHERE CustomerID IS NOT NULL
    GROUP BY CustomerID
),
RFM_SCORED AS (
    SELECT
        CustomerID, Recency_Days, Frequency, Monetary,
        NTILE(3) OVER (ORDER BY Recency_Days DESC) AS r_score,
        NTILE(3) OVER (ORDER BY Frequency ASC) AS f_score,
        NTILE(3) OVER (ORDER BY Monetary ASC) AS m_score
    FROM RFM_BASE
)
SELECT CustomerID, Recency_Days, Frequency, Monetary,
       r_score, f_score, m_score,
       (r_score + f_score + m_score) AS RFM_total,
       CASE
           WHEN (r_score + f_score + m_score) = 9 THEN 'Champions'
           WHEN (r_score + f_score + m_score) >= 7 THEN 'Loyal Customers'
           WHEN (r_score + f_score + m_score) >= 5 THEN 'At Risk'
           ELSE 'Lost'
       END AS RFM_segment
FROM RFM_SCORED
ORDER BY RFM_total DESC;
```

Sample Results — Top Champions

CustomerID	Recency (days)	Frequency	Monetary (£)	R	F	M	Total	Segment
12528	34	4	78.30	3	3	3	9	Champions
14698	18	2	78.54	3	3	3	9	Champions
14432	18	3	78.62	3	3	3	9	Champions
14710	11	2	79.38	3	3	3	9	Champions
17652	31	4	79.60	3	3	3	9	Champions



KEY INSIGHT: The RFM analysis used NTILE(3), a SQL window function that automatically divides customers into 3 equal groups for each dimension rather than using manually set thresholds. Champions scored 9/9: bought recently, buy often, and spend the most. This is more statistically robust than manual threshold setting.



RECOMMENDATION: Create a VIP programme for Champions: early access to new products, exclusive discounts. Identify all 'Lost' customers (score 3) and attempt win-back campaigns with a special offer. Monitor 'At Risk' customers monthly, a declining RFM score is an early warning sign before they churn.

Section 7: Cohort Retention Analysis — The True Funnel

Business Question: Of customers who first purchased in each month, what % returned to buy again in subsequent months? This is the most powerful funnel analysis — showing exactly where and when customers drop off.

SQL Query — Cohort Index Building

```
-- Step 1: Find each customer's first purchase month (cohort assignment)
SELECT CustomerID,
       DATE_FORMAT(MIN(Date), '%Y-%m') AS cohort_month
FROM `e-commerce_clean`
GROUP BY CustomerID;

-- Step 2: Map every customer's activity month to their cohort
WITH first_purchase AS (
  SELECT CustomerID,
         DATE_FORMAT(MIN(Date), '%Y-%m-01') AS cohort_month
  FROM `e-commerce_clean`
  GROUP BY CustomerID
)
SELECT DISTINCT
  `e-commerce_clean`.CustomerID,
  DATE_FORMAT(`e-commerce_clean`.Date, '%Y-%m-01') AS activity_month,
  first_purchase.cohort_month
FROM `e-commerce_clean`
JOIN first_purchase
  ON `e-commerce_clean`.CustomerID = first_purchase.CustomerID;
```

Cohort Index Results (Sample)

Cohort Month	Month Number	Active Customers
2010-12-01	0 (acquisition month)	398
2010-12-01	1 (1 month later)	97
2010-12-01	2	91
2010-12-01	3	97
2010-12-01	4	104
2010-12-01	5	101
2010-12-01	6	99
2010-12-01	11	60
2011-01-01	0	247
2011-01-01	1	46
2011-01-01	2	62



KEY INSIGHT: The Dec-2010 cohort started with 398 customers. By Month 1 only 97 remained active, a 75.6% drop-off after the first purchase! However customers who survived to Month 3-6 showed strong retention (97-104 active) suggesting once customers make 3+ purchases they become loyalists. The Jan-2011 cohort shows an even sharper drop from 247 to just 46 in Month 1 (81.4% drop-off). This data will be visualised as a heatmap in Power BI.

💡 **RECOMMENDATION:** The biggest opportunity is converting Month-0 customers to Month-1 buyers. A targeted follow-up email or offer within 2 weeks of first purchase could dramatically improve Month-1 retention. Even improving Month-1 retention by 10% for the Dec cohort would retain 40 more customers worth an estimated £8,000+ in lifetime revenue.

SQL Concepts Demonstrated

SQL Concept	Where Used	Purpose
LAG() Window Function	Section 2 — Monthly Revenue	Compare each month to the previous month for MoM growth %
NTILE() Window Function	Section 6 — RFM Scoring	Divide customers into 3 equal tiers automatically
DATEDIFF()	Section 6 — RFM Recency	Calculate days since last purchase vs dataset max date
CASE WHEN / END	Sections 4 & 6	Categorise customers into segments based on conditions
CTE — WITH clause	Sections 4, 6, 7	Multi-step queries broken into readable logical blocks
SUM() OVER()	Section 4	Calculate % of total revenue per segment
COUNT(DISTINCT)	Sections 4, 5, 6	Count unique invoices per customer accurately
DATE_FORMAT()	Section 7 — Cohort	Extract and format month from date for cohort grouping
MIN(Date) with GROUP BY	Section 7	Find each customer's first purchase date
JOIN	Section 7 — Cohort	Map customer activity to their acquisition cohort
LIMIT	Section 3 — Products	Filter to Top 10 and Bottom 10 products
ORDER BY ASC/DESC	All sections	Sort results for meaningful ranked output
ROUND()	All sections	Clean numeric formatting to 2 decimal places

This SQL analysis was performed independently on a real-world dataset of 392,733 transactions using MySQL Workbench 8.0. The analysis covers the full customer funnel from acquisition and product performance through to retention, RFM segmentation, and cohort analysis using industry-standard SQL techniques.